1 Generated reference tables

The tables in this appendix are generated from the source tree by `scripts/gen_report_tables.py`. They are not maintained by hand: a renamed flag, a new cycle-slip detector, or a changed state-key delimiter appears here on the next run of the generator. Run the generator with `--check` in CI to fail the build if the committed tables are stale.

1.1 Processing modes

Table 1: `ProcessingMode` – the estimation algorithm selector.

Value	Meaning
`PPP`	Precise Point Positioning (single receiver, external products).
`PPP_RTK`	PPP with RTK enhancements (single receiver, uncombined).
`BaselineSingle`	Double-differenced single baseline: one rover + one reference station.
`BaselineMulti`	Double-differenced multi-receiver network (star topology: N rovers against one reference).
`PPP_RTK_Network`	Minimum-constraint S-system network PRODUCER: joint multi-station estimator emitting satellite clock + code/phase bias (+ atmosphere) under a declared S-basis.
`PPP_RTK_User`	PPP-RTK CONSUMER of network PPP_RTK_Network products (post-processing closed-loop counterpart): uncombined PPP-RTK + network SSR bias/atmosphere consume + S-basis cross-check.

1.2 Execution mode and data source

Table 2: `AppMode` – top-level post-processing vs. real-time dispatch.

Value	Meaning
`PostProcessing`	Batch processing from RINEX / product files.
`RealTime`	Streaming / live processing from RTCM data.

Table 3: `DataSource` – bundled fixtures vs. auto-download.

Value	Meaning
`Internal`	Use files under internal_data_root only.
`Production`	Auto-download from CDDIS into data_root.

1.3 Differencing regime

Table 4: `DifferencingMode` – the measurement differencing regime.

Value	Meaning
`None`	Raw observables (PPP / PPP-RTK without SD).
`BetweenSatellite`	Per-constellation single difference vs. pivot (PPP-RTK SD).
`DoubleDifference`	Between-receiver + between-satellite (baseline processing).

1.4 Config-to-differencing mapping

The differencing regime is resolved once per run by the factory below (verbatim from `MeasurementBuilder.cpp`):

```
DifferencingMode differencingModeFor(const GnssConfig& config) {
    // The network producer runs on undifferenced + uncombined rows and estimates
    // the satellite-side bias lumps explicitly -- it must NEVER apply receiver-side
    // BSSD even if a between_satellite_sd toggle is present in its config.
    if (config.getProcessingMode() == ProcessingMode::PPP_RTK_Network)
                                        return DifferencingMode::None;
    if (config.isBaselineMode())        return DifferencingMode::DoubleDifference;
    if (config.enableBetweenSatelliteSd()) return DifferencingMode::BetweenSatellite;
    return DifferencingMode::None;
}
```

1.5 Feature toggles

These are the loop- and state-vector-level switches.

Table 5: Top-level boolean feature toggles, parsed from the `jval<bool>` calls in `GnssConfigLoad.cpp`. The legacy alias is the pre-refactor flat key still accepted as a fallback.

JSON path	Default	Legacy alias
`estimation.accumulate_phase_windup`	off	`accumulate_phase_windup`
`estimation.adaptive_iono_process_noise.trace`	off	`adaptive_iono_q_trace`
`estimation.aiding.baro`	off	`aiding_baro`
`estimation.aiding.nhc`	off	`aiding_nhc`
`estimation.aiding.vision`	off	`aiding_vision`
`estimation.aiding.wheel`	off	`aiding_wheel`
`estimation.aiding.zupt`	off	`aiding_zupt`
`estimation.ambiguity_resolution.enable_partial_ar`	off	`ar_enable_partial_ar`
`estimation.ambiguity_resolution.wl_only`	off	`ar_wl_only`

JSON path	Default	Legacy alias
estimation.between_satellite_sd.enable_glonass	off	enable_glonass_bssd
estimation.between_satellite_sd.enabled	off	enable_between_satellite_sd
estimation.cycle_slip.ewl	off	cycle_slip_ewl
estimation.cycle_slip.gf	off	cycle_slip_gf
estimation.cycle_slip.graphic	off	cycle_slip_graphic
estimation.cycle_slip.mw	on	cycle_slip_mw
estimation.cycle_slip.turbo_edit	off	cycle_slip_turbo_edit
estimation.cycle_slip.turbo_edit_bridge_lli	off	cycle_slip_turbo_edit_bridge_lli
estimation.cycle_slip.turbo_edit_mw_only_veto	on	cycle_slip_turbo_edit_mw_only_veto
estimation.defer_until_corrections_ready.drop_broadcast_grade	off	defer_until_corrections_ready_drop_broadcast_grade
estimation.defer_until_corrections_ready.require_all_above_mask	off	defer_until_corrections_ready_require_all_above_mask
estimation.enable_cio_frame	off	enable_cio_frame
estimation.enable_data_snooping	off	enable_data_snooping
estimation.enable_full_r_kernel	off	enable_full_r_kernel
estimation.enable_huber_innovation	off	enable_huber_innovation
estimation.enable_huber_r_inflation	on	enable_huber_r_inflation
estimation.enable_if_slip_detector	off	enable_if_slip_detector
estimation.enable_subcycle_slip_suppression	on	enable_subcycle_slip_suppression
estimation.enable_variance_factor	off	enable_variance_factor
estimation.estimate_ifb	off	estimate_ifb
estimation.estimate_phase_clock	off	estimate_phase_clock
estimation.estimate_sat_biases	off	estimate_sat_biases
estimation.estimate_sat_phase_biases	off	estimate_sat_phase_biases
estimation.estimate_tropo_gradients	off	estimate_tropo_gradients
estimation.estimate_velocity	off	estimate_velocity
estimation.glonass_ifb.enable	off	enable_glonass_ifb
estimation.lock_signals_to_clock_ref	off	lock_signals_to_clock_ref
estimation.promote_sat_code_bias	off	promote_sat_code_bias
estimation.require_bia_for_signals	off	require_bia_for_signals
estimation.residual_history.enabled	off	residual_history_enabled
processing.auto_report	off	auto_report
processing.enable_neq_export	off	enable_neq_export
processing.enable_noncausal_slip_bridge	off	enable_noncausal_slip_bridge
processing.enable_obs_skip_to_latest	off	enable_obs_skip_to_latest
processing.enable_obs_ssr_sync	off	enable_obs_ssr_sync
processing.enable_orbit_normal_attitude	off	enable_orbit_normal_attitude

JSON path	Default	Legacy alias
processing.enable_trig_sp3_interp	off	enable_trig_sp3_interp
processing.exclude_eclipsing_satellites	off	exclude_eclipsing_satellites
processing.inhouse_bias_estimation	off	inhouse_bias_estimation
processing.relinearize_geometry	off	relinearize_geometry
processing.report_async	off	report_async
processing.run_smoother	off	run_smoother
processing.uncombined_enabled	off	uncombined_enabled
receiver.enable_antenna_orientation	off	enable_antenna_orientation
receiver.enable_antenna_substitution	off	enable_antenna_substitution
receiver.enable_itrf_velocity	off	enable_itrf_velocity

1.6 Model-correction switches

Table 6: Physical model on/off switches (ModelCorrections). These gate the correction terms applied to the undifferenced observation model.

Field	Default	Effect
enable_troposphere	on	Saastamoinen ZHD + GMF dry mapping (slant hydrostatic delay).
enable_klobuchar_ionosphere	off	Klobuchar ionospheric delay model (single-frequency fallback only). IF combinations and DD short-baselines cancel ionosphere geometrically; PPP-RTK uncombined estimates per-satellite slant iono states. In all three of those configs the Klobuchar model is a no-op — but constructing the Ionosphere instance still pays for an extra Model::Geometry load and the Klobuchar coefficient parse. Default off. Enable only for single-frequency PPP runs that need a-priori iono.

Field	Default	Effect
enable_higher_order_ionosphere	off	Apply second-order (B·cos(θ)·STEC/f^3) and third-order (η·STEC^2/f^4) ionospheric delays as a per-signal observation-level correction, evaluated at the GIM-derived IPP using a centered tilted dipole geomagnetic field. Requires a loaded IONEX GIM as the absolute-STEC source; the engine auto-loads the GIM when this flag is on even if estimation.ionosphere_weighting.enabled is false. No-op when no GIM resolves (warning logged at startup). Magnitudes: I_2 ˜1–2 cm at L1 during high TEC, I_3 ˜mm; the IF combination only partially cancels these, so the largest visible benefit is in standard PPP (IF). For PPP-RTK uncombined and short-baseline DD the higher-order term mostly cancels in the SD / DD differencing, so this flag is closer to a consistency knob than a bias remover. Default off.
enable_relativistic_correction	on	Periodic relativistic clock correction (-2 rv / c).
enable_shapiro_delay	on	Shapiro (gravitational path) delay.
enable_solid_earth_tide	on	IERS 2010 solid Earth tide displacement.
enable_ocean_tide_loading	on	Ocean tidal loading displacement (BLQ).
enable_pole_tide	on	Pole tide displacement (from ERP): IERS 2010 Sec. 7.1.4 rotational deformation.
enable_ocean_pole_tide	off	Ocean pole tide load displacement (IERS 2010 Sec. 7.1.5 / Desai 2002), added on top of enable_pole_tide. Requires the opole load grid (files.opt_grid_path or ¡data¿/static/opole/). ˜0.1-2 mm; default off keeps runs byte-identical.
enable_atmospheric_loading	on	Atmospheric pressure loading displacement.
enable_satellite_pco	on	Satellite antenna phase centre offset.
enable_satellite_pcv	on	Satellite antenna phase centre variation.
enable_receiver_pco	on	Receiver antenna phase centre offset.
enable_receiver_pcv	on	Receiver antenna phase centre variation.
enable_receiver_cross_system_atx_fallback	off	RECEIVER-ONLY: when the receiver ATX entry lacks a non-GPS frequency, substitute the nearest GPS band (E1-¿G01, E5a-¿G05, B1C-¿G01, B2a-¿G05, ...). Valid: receiver phase pattern is frequency-dependent on shared hardware. NEVER applied to the satellite antenna (E01 and G01 are different spacecraft). Opt-in.

Field	Default	Effect
enable_atx_frequency_fallback	on	When the ATX entry doesn't have PCO/PCV for the requested frequency (common for L5 / E5* / B2a on older satellites or receiver antennas), look up the nearest available frequency on the same antenna and use its values instead of silently skipping the correction. Cross-frequency PCO drift on a single physical antenna is typically ¡1 cm — better than the few-cm vertical residual that comes from skipping a frequency entirely. Set false to revert to the legacy "skip if missing" behavior for A/B comparison.
require_atx_coverage	off	When true, drop observations whose (sat, signal) lacks complete ATX coverage on **both** the satellite and receiver entries — i.e., satellite PCO+PCV and receiver PCO+PCV must all resolve to non-empty entries for the row's exact frequency, *without* enable_atx_frequency_fallback being used to substitute a neighbouring frequency. Default false (legacy behaviour: observations are kept and any missing PCO/PCV is silently zeroed). Set true when you've explicitly chosen 'signal_mode: "all"' and want the ATX file to be the authoritative whitelist, dropping rows that would otherwise process without proper antenna calibration.
require_atx_coverage_for_ar	off	When true, AR (WL constraints, LAMBDA NL search) skips (sat, signal) pairs lacking strict ATX coverage even though the underlying observations are still used in the float Kalman update. Softer counterpart to require_atx_coverage: lets you keep "all"-signal float estimation alive while restricting integer fixing to the ATX-curated subset. Default false (no AR-side gating).
restrict_ar_for_substituted_atx	off	When true, AR additionally skips (sat, signal) pairs whose ATX correction was computed via the frequency-fallback substitution chain (rather than an exact entry). Targets specifically the cases where the antenna model is approximate; signals lacking ATX entirely (no substitution applied) are still allowed. Use alongside or instead of require_atx_coverage_for_ar depending on which kind of imperfection you want to gate on. Default false.
loosen_amb_q_for_substituted_atx	off	When true, ambiguity states whose underlying (sat, signal) had their ATX correction frequency-fallback-substituted use a looser process noise (ambiguity_q_substituted_atx) instead of the standard process_noise.ambiguity. Compensates for the small but non-zero modeling error introduced by the substituted PCV/PCO. Default false.

Field	Default	Effect
apply_phase_shifts	off	When true, RINEX phase observations have the receiver-applied "SYS / PHASE SHIFT" values subtracted at parse time, recovering the unshifted phase convention. Default false because most modern BIA products (CODE, CAS, GFZ) are calibrated against the receiver-shifted phases as recorded in RINEX, so leaving the shifts in place is the standard PPP convention. Set true only when your bias product is explicitly calibrated for the unshifted convention, or when running without external BIA and using in-house bias estimation that benefits from a uniform shift-free input.
enable_phase_windup	on	Carrier-phase wind-up correction.
enable_code_phase_bias	on	Code/phase bias corrections from BIA product files.
refine_emission_time	on	When true, Geometry::computeSatelliteGeometry runs a

1.7 Logging and trace channels

Table 7: Logging and CSV-trace channels (LoggingConfig). Each toggles one spdlog file logger or one per-epoch CSV trace.

Field	Default	Channel
enable_main_log	on	logs/main_log.log
enable_gnss_config	on	logs/gnss_config.log
enable_kalman_filter	on	logs/kalman_filter.log
enable_smoother	on	logs/smoother.log
enable_state_manager	on	logs/state_manager.log
enable_ppp_model	on	logs/ppp_model.log
enable_measurement_model	on	logs/measurement_model.log
enable_mw_slip_detector	on	logs/mw_slip_detector.log
enable_if_slip_detector	on	logs/if_slip_detector.log
enable_gf_slip_detector	on	logs/gf_slip_detector.log
enable_turbo_edit_resolver	on	logs/turbo_edit_resolver.log
enable_cycle_slip_manager	on	logs/cycle_slip_manager.log
enable_ambiguity_resolver	on	logs/ambiguity_resolver.log
enable_corrections	on	logs/corrections.log
enable_geometry	on	logs/geometry.log
enable_troposphere	on	logs/troposphere.log

Field	Default	Channel
enable_ionosphere	on	logs/ionosphere.log
enable_antenna	on	logs/antenna.log
enable_antenna_model	on	logs/antenna_model.log
enable_ocean_tide	on	logs/ocean_tide.log
enable_planetary_model	on	logs/planetary_model.log
enable_stochastic_model	on	logs/stochastic_model.log
enable_rinex_parser	on	logs/rinex_parser.log
enable_product_parser	on	logs/product_parser.log
enable_product_cache	on	logs/product_cache.log
enable_resolver	on	logs/resolver.log
enable_state_initializer	on	logs/state_initializer.log
enable_bssd_pivot_manager	on	logs/bssd_pivot_manager.log
enable_bssd_transform	on	logs/bssd_transform.log
enable_ppp_utils	on	logs/ppp_utils.log
enable_downloader	on	logs/downloader.log
enable_ntrip	on	logs/ntrip.log
enable_rt_engine	on	logs/rt_engine.log
enable_imu	on	logs/imu.log
enable_state_trace	on	logs/state_trace.csv
enable_ambiguity_trace	on	logs/ambiguity_trace.csv
enable_ionosphere_trace	on	logs/ionosphere_trace.csv
enable_sat_bias_trace	on	logs/sat_bias_trace.csv
enable_sat_status_trace	on	logs/satellite_status_trace.csv
enable_dop_trace	on	logs/dop_trace.csv
enable_residual_trace	on	logs/measurement_residuals.csv
enable_wls_trace	on	logs/wls_trace.csv
enable_smoother_trace	on	logs/smoother_trace.csv
enable_mw_slip_trace	on	logs/mwwl_trace.csv
enable_if_slip_trace	on	logs/if_slip_trace.csv
enable_gf_slip_trace	on	logs/gf_slip_trace.csv
enable_turbo_edit_trace	on	logs/turbo_edit_trace.csv
enable_windup_trace	off	logs/windup_trace.csv (phase wind-up diagnostic; off by default)
kalman_filter_concise_log	off	

1.8 Cycle-slip detector registry

Table 8: Cycle-slip detectors known to `SlipDetectorRegistry`. *Active* detectors have a creator in the factory table and run when their flag is set; *staged* detectors are declared in the `SlipDetectorKind` enum but excluded from the build (the registry warns and skips them if requested). The $k\sigma$ column is the MAD gate multiplier from `CycleSlipConfig`.

Detector	Config flag	Default	$k\sigma$	Status	Combination
`MelbourneWubbena`	`estimation.cycle_slip.mw`	on	8.0	**active**	Wide-lane MW combination (primary, always available).
`Graphic`	`estimation.cycle_slip.graphic`	off	6.0	**active**	GRAPHIC (code+phase average) per-signal detector.
`ExtraWideLane`	`estimation.cycle_slip.ewl`	off	5.0	**active**	Extra-wide-lane MW for 3+ frequency satellites.
`GeometryFree`	`estimation.cycle_slip.gf`	off	–	*staged*	2nd-order time-differenced geometry-free (STAGED).
`IonoFreePhase`	`estimation.cycle_slip.ifp`	off	–	*staged*	Double-differenced ionosphere-free phase OMC (STAGED).

1.9 State-vector key schema

Table 9: Dynamic state-vector key formats, reconstructed from the `make*Key` bodies in `StateKeySchema.hpp`. Angle-bracketed fields are runtime values; literal characters are the delimiter / infix / prefix / suffix constants.

State	Key format	Builder
Ambiguity	`<satellite>_<signal>`	`makeAmbiguityKey`
NL Ambiguity	`<satellite>_NL_<band>`	`makeNlAmbiguityKey`
WL Ambiguity	`<satellite>_WL_<band_a>_<band_b>`	`makeWlAmbiguityKey`
DD Ambiguity	`DD|<cons>|<sat_i>|<code>|<pivot>`	`makeDdAmbiguityKey`
Ionosphere	`<satellite>`	`makeIonosphereKey`
IFB	`<system>_IFB<band>`	`makeIfbKey`
DCB	`<system>_DCB`	`makeDcbKey`
Glonass Per Slot IFB	`<system>_IFB<band>_k`	`makeGlonassPerSlotIfbKey`
Satellite Bias	`<satellite>|<signal>`	`makeSatBiasKey`

State	Key format	Builder
Satellite Phase Bias	`<satellite>|<signal>|PB`	`makeSatPhaseBiasKey`
System Time Offset	`<system>`	`makeSystemTimeOffsetKey`
Pivot Seed	`<cons>|<band>`	`makePivotSeedKey`